# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## ANSWER KEY & MARKING SCHEME SET 09

**Full Syllabus (Chapters 1–13)** | Class 9 Science | Maximum Marks: 80

### SECTION — A (Objective Type) — Answers — 20 × 1 = 20

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** describes a regular pattern observed in nature | laws describe what happens |
| 2 | **(c)** local objects or opinions | fairness in trade and research |
| 3 | **(d)** soda water ($CO_2$ in water) | uniform composition throughout |
| 4 | **(c)** larger, well-formed crystals | particles get time to arrange |
| 5 | **(c)** 95 u | 24 + 71 |
| 6 | **(c)** 1 : 8 | Law of constant proportions |
| 7 | **(d)** 2, 8, 8, 2 | fill shells in order with maximum 2, 8, 8, 2 |
| 8 | **(c)** 56 m | $s = ut + \frac{1}{2}at^2$ |
| 9 | **(a)** velocity | rate of change of position |
| 10 | **(c)** 1.5 m $s^{-2}$ | a = F/m |
| 11 | **(c)** smaller acceleration | $a \propto 1/m$ |
| 12 | **(d)** the fulcrum is between effort and load | first class |
| 13 | **(d)** timbre | overtone pattern |
| 14 | **(a)** programmed cell death | PCD |
| 15 | **(c)** 0.2 mm (200 µm) | 5 mm ÷ 25 |
| 16 | **(c)** squamous epithelium | gas exchange |
| 17 | **(b)** Monera | prokaryotes |
| 18 | **(c)** ovary | each has an egg cell |
| 19 | **(d)** Louis Pasteur | germ theory |
| 20 | **(c)** 10 kWh | energy = power × area × time |

### SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Matter: mass of ball, speed and direction of the hit. Can be ignored: brand of bat, colour of ball, grass on the field (1+1). | 1+1 |
| 22 | H : O = 1 : 8, so 1 g H and 8 g O (2). | 1+1 |
| 23 | Distance = 3 + 4 = 7 m (1); displacement = $\sqrt{(3^2 + 4^2)} = 5$ m (1). | 1+1 |
| 24 | They act on two different objects (2). | 2 |
| 25 | The rigid cell wall resists the pressure from absorbed water (2). | 2 |
| 26 | A group of similar cells performing a specific function (1); increases efficiency (1). | 1+1 |
| 27 | Thread-like structures in the nucleus carrying genetic information (1); 46 (23 pairs) (1). | 1+1 |
| 28 | Geosphere, hydrosphere, atmosphere and biosphere (cryosphere may be mentioned) (2). | 2 |

### SECTION — C (Short Answer) — Answers — 8 × 3 = 24

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 29 | Layers form because of a density difference; with equal densities there is no force separating them, so they stay mixed as droplets (3). | 1+1+1 |
| 30 | In the modern model electrons exist as electron clouds; only the regions of highest probability can be predicted (3). | 1+1+1 |
| 31 | KE is due to motion, e.g. a rolling ball (1.5); PE is due to position or shape, e.g. a stretched spring (1.5). | 1½+1½ |
| 32 | Sound makes the eardrum vibrate (1); small bones amplify the vibrations (1); the cochlea converts them into electrical signals sent to the brain (1). | 1+1+1 |
| 33 | Ribs are attached by flexible cartilage (1); they move to expand and contract the chest (1); so air moves in and out of the lungs (1). | 1+1+1 |

| | | |
|---|---|---|
| **34** | Segmented body; organ system level; muscles for movement; e.g. earthworm (1+1+1). | 1+1+1 |
| **35** | O (2,6) needs two electrons; each shares two electrons (2); two shared pairs give a double bond O=O (1). | 2+1 |
| | **OR:** Cl $= \frac{35.5}{23} \times 46 = 71$ g (2); NaCl $= 46 + 71 = 117$ g (1). | 2+1 |
| **36** | CFCs release chlorine that destroys ozone (1.5); the Montreal Protocol banned CFCs, so the layer is slowly recovering (1.5). | 1½+1½ |
| | **OR:** Enhanced greenhouse effect (1); global warming, glacier melting, sea-level rise, extreme weather (any two) (2). | 1+2 |

## SECTION — D (Long Answer) — Answers

**4 × 5 = 20**

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| **37** | (i) NaCl solution (1). (ii) It forms free ions $Na^+$ and $Cl^-$; sugar does not form ions (2). (iii) NaCl has a high melting point; sugar has low (1). (iv) NaCl ionic; sugar covalent (1). | 1+2+1+1 |
| | **OR:** (i) 12 : 32 = 3 : 8 (1). (ii) $\frac{12}{44} \times 22 = 6$ g (2). (iii) $\frac{8}{3} \times 6 = 16$ g (2). | 1+2+2 |
| **38** | (i) Mass is fixed, weight depends on g (2). (ii) 600 N (1.5). (iii) 96 N (1.5). | 2+1½+1½ |
| | **OR:** (i) First: lurch in a bus; second: F = ma for a car; third: rocket launch (3). (ii) First law (inertia) – the body continues moving forward (2). | 3+2 |
| **39** | (i) Biconcave, no nucleus, full of haemoglobin (2). (ii) Nerve cell, muscle cell or root hair cell with its function (2). (iii) About 120 days (1). | 2+2+1 |
| | **OR:** (i) Growing cells outside the body (1). (ii) To prevent contamination by microbes (1). (iii) Production of biochemicals or study of cells (1). (iv) The ability of a plant cell to grow into a whole plant (2). | 1+1+1+2 |
| **40** | (i) It forms gametes with half the chromosome number (1). (ii) 23, 23 and 46 (2). (iii) Each gamete receives a random combination of chromosomes from each pair (2). | 1+2+2 |
| | **OR:** (i) Balanced diet rich in proteins, vitamins and minerals (1). (ii) Complete nutrition and protection from infection (1). (iii) Anxiety or low mood after delivery that can be treated (1). (iv) To prevent infections in mother and baby (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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